

CHEMISTRY SIMS SIUT

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ATOMIC STRUCTURE - 20 MCQs

1. Who discovered the electron?

- A) Rutherford
- B) J.J. Thomson ✓
- C) Chadwick
- D) Goldstein

Answer: B

2. Who discovered the proton?

- A) J.J. Thomson
- B) Rutherford ✓
- C) Chadwick
- D) Goldstein

Answer: B

3. Who discovered the neutron?

- A) J.J. Thomson
- B) Rutherford
- C) Chadwick ✓
- D) Bohr

Answer: C

4. The charge on an electron is:

- A) $+1.6 \times 10^{-19} \text{ C}$
- B) $-1.6 \times 10^{-19} \text{ C}$ ✓
- C) 0

D) $+9.1 \times 10^{-31} \text{ C}$

Answer: B

5. The mass of a proton is approximately:

A) $9.1 \times 10^{-31} \text{ kg}$

B) $1.67 \times 10^{-27} \text{ kg}$ ✓

C) $1.67 \times 10^{-31} \text{ kg}$

D) $9.1 \times 10^{-27} \text{ kg}$

Answer: B

6. The mass of an electron is approximately:

A) $9.1 \times 10^{-31} \text{ kg}$ ✓

B) $1.67 \times 10^{-27} \text{ kg}$

C) $1.67 \times 10^{-31} \text{ kg}$

D) $9.1 \times 10^{-27} \text{ kg}$

Answer: A

7. The region around the nucleus where electrons are found is called:

A) Orbit

B) Orbital ✓

C) Shell

D) Subshell

Answer: B

8. The maximum number of electrons in the first shell (K shell) is:

A) 2 ✓

B) 8

C) 18

D) 32

Answer: A

9. The maximum number of electrons in the second shell (L shell) is:

- A) 2
- B) 8 ✓
- C) 18
- D) 32

Answer: B

10. The maximum number of electrons in the third shell (M shell) is:

- A) 2
- B) 8
- C) 18 ✓
- D) 32

Answer: C

11. The general formula for maximum electrons in a shell is:

- A) n
- B) n^2
- C) $2n^2$ ✓
- D) $2n$

Answer: C

12. An atom consists of:

- A) Proton and neutron only
- B) Electron and proton only
- C) Proton, neutron and electron ✓
- D) Neutron and electron only

Answer: C

13. The number of protons in an atom is called:

- A) Mass number

- B) Atomic number ✓
- C) Atomic mass
- D) Valency

Answer: B

14. The sum of protons and neutrons in an atom is called:

- A) Atomic number
- B) Mass number ✓
- C) Atomic mass
- D) Valency

Answer: B

15. Isotopes have the same number of:

- A) Neutrons
- B) Protons ✓
- C) Electrons
- D) Mass number

Answer: B

16. Isotopes have different numbers of:

- A) Protons
- B) Electrons
- C) Neutrons ✓
- D) Atomic number

Answer: C

17. The number of orbitals in a subshell is given by:

- A) $2l + 1$ ✓
- B) $2n^2$
- C) $n + l$
- D) $2n$

Answer: A

18. The s-subshell can hold a maximum of:

- A) 2 electrons ✓
- B) 6 electrons
- C) 10 electrons
- D) 14 electrons

Answer: A

19. The p-subshell can hold a maximum of:

- A) 2 electrons
- B) 6 electrons ✓
- C) 10 electrons
- D) 14 electrons

Answer: B

20. The electronic configuration of oxygen (Atomic number 8) is:

- A) $1s^2 2s^2 2p^4$ ✓
- B) $1s^2 2s^2 2p^6$
- C) $1s^2 2s^2 2p^3$
- D) $1s^2 2s^1 2p^5$

Answer: A

PERIODIC TABLE & PERIODICITY - 20 MCQs

1. The periodic table was first proposed by:

- A) Mendeleev ✓
- B) Newlands
- C) Moseley

D) Döbereiner

Answer: A

2. The modern periodic table is based on:

- A) Atomic mass
- B) Atomic number ✓
- C) Valency
- D) Atomic radius

Answer: B

3. The periodic table has how many periods?

- A) 7 ✓
- B) 8
- C) 18
- D) 32

Answer: A

4. The periodic table has how many groups?

- A) 7
- B) 8
- C) 18 ✓
- D) 32

Answer: C

5. Elements in the same group have the same:

- A) Number of shells
- B) Valence electrons ✓
- C) Atomic number
- D) Atomic mass

Answer: B

6. Elements in the same period have the same:

- A) Number of shells ✓
- B) Valence electrons
- C) Atomic number
- D) Atomic mass

Answer: A

7. Across a period from left to right, atomic radius:

- A) Increases
- B) Decreases ✓
- C) Remains same
- D) First increases then decreases

Answer: B

8. Down a group, atomic radius:

- A) Increases ✓
- B) Decreases
- C) Remains same
- D) First increases then decreases

Answer: A

9. Ionization energy across a period from left to right:

- A) Increases ✓
- B) Decreases
- C) Remains same
- D) First increases then decreases

Answer: A

10. Ionization energy down a group:

- A) Increases

- B) Decreases ✓
- C) Remains same
- D) First increases then decreases

Answer: B

11. Electronegativity across a period from left to right:

- A) Increases ✓
- B) Decreases
- C) Remains same
- D) First increases then decreases

Answer: A

12. Electronegativity down a group:

- A) Increases
- B) Decreases ✓
- C) Remains same
- D) First increases then decreases

Answer: B

13. The most electronegative element is:

- A) Oxygen
- B) Nitrogen
- C) Fluorine ✓
- D) Chlorine

Answer: C

14. The least electronegative element is:

- A) Hydrogen
- B) Helium
- C) Cesium ✓
- D) Lithium

Answer: C

15. Elements in group 1 are called:

- A) Alkali metals ✓
- B) Alkaline earth metals
- C) Halogens
- D) Noble gases

Answer: A

16. Elements in group 2 are called:

- A) Alkali metals
- B) Alkaline earth metals ✓
- C) Halogens
- D) Noble gases

Answer: B

17. Elements in group 17 are called:

- A) Alkali metals
- B) Alkaline earth metals
- C) Halogens ✓
- D) Noble gases

Answer: C

18. Elements in group 18 are called:

- A) Alkali metals
- B) Alkaline earth metals
- C) Halogens
- D) Noble gases ✓

Answer: D

19. Which of the following is a metalloid?

- A) Sodium
- B) Silicon ✓
- C) Chlorine
- D) Argon

Answer: B

20. The element with the highest ionization energy is:

- A) Fluorine
- B) Oxygen
- C) Neon ✓
- D) Helium

Answer: C

CHEMICAL BONDING - 20 MCQs

1. The force that holds atoms together in a compound is called:

- A) Chemical bond ✓
- B) Physical bond
- C) Hydrogen bond
- D) Van der Waals force

Answer: A

2. A bond formed by the transfer of electrons is called:

- A) Covalent bond
- B) Ionic bond ✓
- C) Metallic bond
- D) Hydrogen bond

Answer: B

3. A bond formed by the sharing of electrons is called:

- A) Ionic bond
- B) Covalent bond ✓
- C) Metallic bond
- D) Hydrogen bond

Answer: B

4. The bond in NaCl is:

- A) Covalent bond
- B) Ionic bond ✓
- C) Metallic bond
- D) Hydrogen bond

Answer: B

5. The bond in H₂O is:

- A) Ionic bond
- B) Covalent bond ✓
- C) Metallic bond
- D) Hydrogen bond

Answer: B

6. The bond in N₂ is:

- A) Single covalent bond
- B) Double covalent bond
- C) Triple covalent bond ✓
- D) Ionic bond

Answer: C

7. The number of bonds in oxygen molecule (O₂) is:

- A) 1

- B) 2 ✓
- C) 3
- D) 4

Answer: B

8. The number of bonds in nitrogen molecule (N_2) is:

- A) 1
- B) 2
- C) 3 ✓
- D) 4

Answer: C

9. The bond in H_2 is:

- A) Single covalent bond ✓
- B) Double covalent bond
- C) Triple covalent bond
- D) Ionic bond

Answer: A

10. The shape of methane (CH_4) is:

- A) Linear
- B) Trigonal planar
- C) Tetrahedral ✓
- D) Bent

Answer: C

11. The shape of water (H_2O) is:

- A) Linear
- B) Trigonal planar
- C) Tetrahedral
- D) Bent ✓

Answer: D

12. The bond angle in methane (CH_4) is:

- A) 90°
- B) 104.5°
- C) 109.5° ✓
- D) 120°

Answer: C

13. The bond angle in water (H_2O) is:

- A) 90°
- B) 104.5° ✓
- C) 109.5°
- D) 120°

Answer: B

14. The bond angle in CO_2 is:

- A) 90°
- B) 104.5°
- C) 109.5°
- D) 180° ✓

Answer: D

15. The type of bonding in metals is:

- A) Ionic bond
- B) Covalent bond
- C) Metallic bond ✓
- D) Hydrogen bond

Answer: C

16. The intermolecular force responsible for the unique properties of water is:

- A) Ionic bond
- B) Covalent bond
- C) Hydrogen bond ✓
- D) Metallic bond

Answer: C

17. VSEPR theory stands for:

- A) Valence Shell Electron Pair Repulsion ✓
- B) Valence State Electron Pair Repulsion
- C) Valence Shell Electron Proton Repulsion
- D) None of these

Answer: A

18. The octet rule states that atoms tend to have:

- A) 2 electrons in the outermost shell
- B) 8 electrons in the outermost shell ✓
- C) 18 electrons in the outermost shell
- D) 32 electrons in the outermost shell

Answer: B

19. The number of valence electrons in carbon is:

- A) 2
- B) 3
- C) 4 ✓
- D) 5

Answer: C

20. The number of valence electrons in chlorine is:

- A) 2
- B) 4
- C) 6

D) 7 ✓

Answer: D

STATES OF MATTER - 20 MCQs

1. The three states of matter are:

- A) Solid, liquid, gas ✓
- B) Solid, liquid, plasma
- C) Liquid, gas, plasma
- D) Solid, gas, plasma

Answer: A

2. The state of matter with a definite shape and volume is:

- A) Solid ✓
- B) Liquid
- C) Gas
- D) Plasma

Answer: A

3. The state of matter with a definite volume but no definite shape is:

- A) Solid
- B) Liquid ✓
- C) Gas
- D) Plasma

Answer: B

4. The state of matter with no definite shape or volume is:

- A) Solid

- B) Liquid
- C) Gas ✓
- D) Plasma

Answer: C

5. The particles in a solid are:

- A) Closely packed ✓
- B) Loosely packed
- C) Very far apart
- D) Randomly arranged

Answer: A

6. The particles in a liquid are:

- A) Closely packed
- B) Loosely packed ✓
- C) Very far apart
- D) Fixed positions

Answer: B

7. The particles in a gas are:

- A) Closely packed
- B) Loosely packed
- C) Very far apart ✓
- D) Fixed positions

Answer: C

8. The process of changing from solid to liquid is called:

- A) Melting ✓
- B) Freezing
- C) Boiling
- D) Condensation

Answer: A

9. The process of changing from liquid to solid is called:

- A) Melting
- B) Freezing ✓
- C) Boiling
- D) Condensation

Answer: B

10. The process of changing from liquid to gas is called:

- A) Melting
- B) Freezing
- C) Boiling/Evaporation ✓
- D) Condensation

Answer: C

11. The process of changing from gas to liquid is called:

- A) Melting
- B) Freezing
- C) Boiling
- D) Condensation ✓

Answer: D

12. The process of changing from solid to gas directly is called:

- A) Melting
- B) Sublimation ✓
- C) Condensation
- D) Deposition

Answer: B

13. The process of changing from gas to solid directly is called:

- A) Melting
- B) Sublimation
- C) Condensation
- D) Deposition ✓

Answer: D

14. Boyle's law states that pressure is inversely proportional to:

- A) Temperature
- B) Volume ✓
- C) Number of moles
- D) Mass

Answer: B

15. Charles's law states that volume is directly proportional to:

- A) Pressure
- B) Temperature ✓
- C) Number of moles
- D) Mass

Answer: B

16. Avogadro's law states that volume is directly proportional to:

- A) Pressure
- B) Temperature
- C) Number of moles ✓
- D) Mass

Answer: C

17. The ideal gas equation is:

- A) $PV = nRT$ ✓
- B) $PV = RT$
- C) $P = nRT/V$

D) $V = nRT/P$

Answer: A

18. The value of standard temperature is:

- A) 0°C or 273.15 K ✓
- B) 25°C or 298.15 K
- C) 100°C or 373.15 K
- D) -273.15°C or 0 K

Answer: A

19. The value of standard pressure is:

- A) 1 atm or 760 torr ✓
- B) 2 atm or 1520 torr
- C) 0.5 atm or 380 torr
- D) 10 atm or 7600 torr

Answer: A

20. The kinetic theory of gases assumes that gas particles:

- A) Have no volume
- B) Have no intermolecular forces
- C) Move randomly and collide elastically
- D) All of these ✓

Answer: D

SOLUTIONS & CONCENTRATIONS - 20 MCQs

1. A homogeneous mixture of two or more substances is called:

- A) Compound

- B) Solution ✓
- C) Suspension
- D) Colloid

Answer: B

2. The component of a solution that is present in the largest amount is called:

- A) Solute
- B) Solvent ✓
- C) Solution
- D) Mixture

Answer: B

3. The component of a solution that is present in a smaller amount is called:

- A) Solute ✓
- B) Solvent
- C) Solution
- D) Mixture

Answer: A

4. A solution where the solute is completely dissolved is called:

- A) Saturated solution
- B) Unsaturated solution ✓
- C) Supersaturated solution
- D) Concentrated solution

Answer: B

5. A solution that cannot dissolve any more solute is called:

- A) Saturated solution ✓
- B) Unsaturated solution
- C) Supersaturated solution
- D) Dilute solution

Answer: A

6. A solution that contains more solute than can normally dissolve is called:

- A) Saturated solution
- B) Unsaturated solution
- C) Supersaturated solution ✓
- D) Dilute solution

Answer: C

7. Molarity is defined as:

- A) Moles of solute per liter of solution ✓
- B) Moles of solute per kilogram of solvent
- C) Grams of solute per liter of solution
- D) Grams of solute per 100 mL of solution

Answer: A

8. Molality is defined as:

- A) Moles of solute per liter of solution
- B) Moles of solute per kilogram of solvent ✓
- C) Grams of solute per liter of solution
- D) Grams of solute per 100 mL of solution

Answer: B

9. The unit of molarity is:

- A) mol/kg
- B) mol/L ✓
- C) g/L
- D) g/mL

Answer: B

10. The unit of molality is:

- A) mol/kg ✓
- B) mol/L
- C) g/L
- D) g/mL

Answer: A

11. A solution with a high concentration of solute is called:

- A) Dilute
- B) Concentrated ✓
- C) Saturated
- D) Unsaturated

Answer: B

12. A solution with a low concentration of solute is called:

- A) Dilute ✓
- B) Concentrated
- C) Saturated
- D) Unsaturated

Answer: A

13. Solubility is the amount of solute that can dissolve in a given amount of:

- A) Solute
- B) Solvent ✓
- C) Solution
- D) Mixture

Answer: B

14. Solubility generally increases with:

- A) Decrease in temperature
- B) Increase in temperature ✓
- C) Decrease in pressure

D) Increase in pressure

Answer: B

15. The solubility of gases in liquids generally increases with:

- A) Decrease in temperature
- B) Increase in temperature
- C) Decrease in pressure
- D) Increase in pressure ✓

Answer: D

16. The process of making a solution by dissolving a solute is called:

- A) Precipitation
- B) Dissolution ✓
- C) Crystallization
- D) Filtration

Answer: B

17. The process of separating a solid from a solution by evaporation is called:

- A) Precipitation
- B) Crystallization ✓
- C) Filtration
- D) Distillation

Answer: B

18. A solution where water is the solvent is called:

- A) Aqueous solution ✓
- B) Non-aqueous solution
- C) Saturated solution
- D) Unsaturated solution

Answer: A

19. A solution where the solvent is not water is called:

- A) Aqueous solution
- B) Non-aqueous solution ✓
- C) Saturated solution
- D) Unsaturated solution

Answer: B

20. The concentration of a solution can be expressed as:

- A) Percentage
- B) Parts per million
- C) Molarity
- D) All of these ✓

Answer: D

ACIDS, BASES & pH - 20 MCQs

1. A substance that donates protons (H^+ ions) is called:

- A) Acid ✓
- B) Base
- C) Salt
- D) Neutral

Answer: A

2. A substance that accepts protons (H^+ ions) is called:

- A) Acid
- B) Base ✓
- C) Salt
- D) Neutral

Answer: B

3. According to Arrhenius, an acid is a substance that produces:

- A) H^+ ions in water ✓
- B) OH^- ions in water
- C) Both H^+ and OH^- ions
- D) None of these

Answer: A

4. According to Arrhenius, a base is a substance that produces:

- A) H^+ ions in water
- B) OH^- ions in water ✓
- C) Both H^+ and OH^- ions
- D) None of these

Answer: B

5. The pH scale ranges from:

- A) 0 to 7
- B) 7 to 14
- C) 0 to 14 ✓
- D) 1 to 10

Answer: C

6. A pH of 7 indicates:

- A) Acidic
- B) Basic
- C) Neutral ✓
- D) None of these

Answer: C

7. A pH less than 7 indicates:

- A) Acidic ✓
- B) Basic
- C) Neutral
- D) None of these

Answer: A

8. A pH greater than 7 indicates:

- A) Acidic
- B) Basic ✓
- C) Neutral
- D) None of these

Answer: B

9. The pH of pure water is:

- A) 0
- B) 7 ✓
- C) 14
- D) 10

Answer: B

10. The pH of stomach acid is approximately:

- A) 2 ✓
- B) 7
- C) 10
- D) 12

Answer: A

11. The pH of blood is approximately:

- A) 2
- B) 7.4 ✓
- C) 10

D) 12

Answer: B

12. Acids turn blue litmus:

- A) Red ✓
- B) Blue
- C) Green
- D) Yellow

Answer: A

13. Bases turn red litmus:

- A) Red
- B) Blue ✓
- C) Green
- D) Yellow

Answer: B

14. The reaction between an acid and a base is called:

- A) Combustion
- B) Neutralization ✓
- C) Precipitation
- D) Oxidation

Answer: B

15. The products of neutralization are:

- A) Acid and base
- B) Salt and water ✓
- C) Salt and hydrogen
- D) Water and oxygen

Answer: B

16. The salt formed from HCl and NaOH is:

- A) NaCl ✓
- B) KCl
- C) CaCl_2
- D) MgCl_2

Answer: A

17. Hydrochloric acid is:

- A) HCl ✓
- B) H_2SO_4
- C) HNO_3
- D) H_2CO_3

Answer: A

18. Sulfuric acid is:

- A) HCl
- B) H_2SO_4 ✓
- C) HNO_3
- D) H_2CO_3

Answer: B

19. Nitric acid is:

- A) HCl
- B) H_2SO_4
- C) HNO_3 ✓
- D) H_2CO_3

Answer: C

20. Sodium hydroxide is:

- A) NaCl

- B) NaOH ✓
- C) KOH
- D) Ca(OH)₂

Answer: B

STOICHIOMETRY & CHEMICAL CALCULATIONS - 20 MCQs

1. Stoichiometry is the study of:
- A) Qualitative relationships
 - B) Quantitative relationships ✓
 - C) Structural relationships
 - D) None of these

Answer: B

2. The mole is the SI unit for:
- A) Mass
 - B) Volume
 - C) Amount of substance ✓
 - D) Concentration

Answer: C

3. One mole of any substance contains how many particles?
- A) 6.02×10^{23} ✓
 - B) 6.02×10^{22}
 - C) 6.02×10^{24}
 - D) 6.02×10^{21}

Answer: A

4. Avogadro's number is:

- A) 6.02×10^{23} ✓
- B) 6.02×10^{22}
- C) 6.02×10^{24}
- D) 6.02×10^{21}

Answer: A

5. The molar mass of an element is numerically equal to its:

- A) Atomic mass ✓
- B) Molecular mass
- C) Formula mass
- D) None of these

Answer: A

6. The molar mass of water (H_2O) is approximately:

- A) 16 g/mol
- B) 18 g/mol ✓
- C) 20 g/mol
- D) 22 g/mol

Answer: B

7. The molar mass of NaCl is approximately:

- A) 40 g/mol
- B) 58.5 g/mol ✓
- C) 72 g/mol
- D) 100 g/mol

Answer: B

8. The number of moles in 36 g of water is:

- A) 1 mole
- B) 2 moles ✓
- C) 3 moles

D) 4 moles

Answer: B

9. The percentage composition of an element in a compound is calculated by:

- A) $(\text{Mass of element} / \text{Mass of compound}) \times 100$ ✓
- B) $(\text{Mass of compound} / \text{Mass of element}) \times 100$
- C) $(\text{Volume of element} / \text{Volume of compound}) \times 100$
- D) None of these

Answer: A

10. A balanced chemical equation shows:

- A) Reactants and products
- B) Number of moles of each substance
- C) Conservation of mass
- D) All of these ✓

Answer: D

11. The limiting reactant is the one that:

- A) Is in excess
- B) Is completely consumed first ✓
- C) Has the most moles
- D) Has the least moles

Answer: B

12. The theoretical yield is:

- A) The actual amount produced
- B) The maximum amount that can be produced ✓
- C) The amount lost during reaction
- D) None of these

Answer: B

13. The actual yield is:

- A) The amount produced experimentally ✓
- B) The maximum amount that can be produced
- C) The amount lost during reaction
- D) None of these

Answer: A

14. Percent yield is calculated as:

- A) $(\text{Actual yield} / \text{Theoretical yield}) \times 100$ ✓
- B) $(\text{Theoretical yield} / \text{Actual yield}) \times 100$
- C) $(\text{Actual yield} - \text{Theoretical yield}) \times 100$
- D) None of these

Answer: A

15. The formula for moles is:

- A) $\text{Mass} / \text{Molar mass}$ ✓
- B) $\text{Molar mass} / \text{Mass}$
- C) $\text{Mass} \times \text{Molar mass}$
- D) None of these

Answer: A

16. The number of moles in 22.4 L of any gas at STP is:

- A) 0.5 mole
- B) 1 mole ✓
- C) 2 moles
- D) 3 moles

Answer: B

17. STP stands for:

- A) Standard Temperature and Pressure ✓

- B) Standard Time and Pressure
- C) Standard Temperature and Volume
- D) None of these

Answer: A

18. A chemical reaction that absorbs heat is called:

- A) Exothermic
- B) Endothermic ✓
- C) Isothermic
- D) None of these

Answer: B

19. A chemical reaction that releases heat is called:

- A) Exothermic ✓
- B) Endothermic
- C) Isothermic
- D) None of these

Answer: A

20. The law of conservation of mass states that:

- A) Mass is created
- B) Mass is destroyed
- C) Mass can neither be created nor destroyed ✓
- D) None of these

Answer: C

ORGANIC CHEMISTRY BASICS - 20 MCQs

1. The study of carbon-containing compounds is called:

- A) Inorganic chemistry
- B) Organic chemistry ✓
- C) Physical chemistry
- D) Analytical chemistry

Answer: B

2. Organic compounds contain:

- A) Carbon and hydrogen ✓
- B) Carbon and oxygen
- C) Carbon and nitrogen
- D) Only carbon

Answer: A

3. The valency of carbon is:

- A) 2
- B) 3
- C) 4 ✓
- D) 5

Answer: C

4. The simplest organic compound is:

- A) Ethane
- B) Methane ✓
- C) Propane
- D) Butane

Answer: B

5. The formula of methane is:

- A) CH_3
- B) CH_4 ✓
- C) C_2H_6

D) C_3H_8

Answer: B

6. The type of bond between carbon atoms in organic compounds is:

- A) Ionic bond
- B) Covalent bond ✓
- C) Metallic bond
- D) Hydrogen bond

Answer: B

7. Organic compounds that have the same molecular formula but different structures are called:

- A) Isotopes
- B) Isomers ✓
- C) Isotones
- D) Isobars

Answer: B

8. The process of breaking down large organic molecules into smaller ones is called:

- A) Cracking ✓
- B) Polymerization
- C) Condensation
- D) Hydrolysis

Answer: A

9. The process of combining small organic molecules into large ones is called:

- A) Cracking
- B) Polymerization ✓
- C) Condensation
- D) Hydrolysis

Answer: B

10. The general formula of alkanes is:

- A) C_nH_{2n}
- B) C_nH_{2n+2} ✓
- C) C_nH_{2n-2}
- D) C_nH_{2n+1}

Answer: B

11. The general formula of alkenes is:

- A) C_nH_{2n}
- B) C_nH_{2n+2}
- C) C_nH_{2n-2}
- D) C_nH_{2n+1}

Answer: A

12. The general formula of alkynes is:

- A) C_nH_{2n}
- B) C_nH_{2n+2}
- C) C_nH_{2n-2} ✓
- D) C_nH_{2n+1}

Answer: C

13. A functional group is:

- A) An atom or group of atoms that determines the properties of a compound ✓
- B) A type of bond
- C) A type of isomer
- D) None of these

Answer: A

14. The functional group of alcohols is:

- A) -OH ✓

- B) -COOH
- C) -CHO
- D) -NH₂

Answer: A

15. The functional group of carboxylic acids is:

- A) -OH
- B) -COOH ✓
- C) -CHO
- D) -NH₂

Answer: B

16. The functional group of aldehydes is:

- A) -OH
- B) -COOH
- C) -CHO ✓
- D) -NH₂

Answer: C

17. The functional group of ketones is:

- A) -OH
- B) -COOH
- C) -CHO
- D) -CO- ✓

Answer: D

18. The functional group of amines is:

- A) -OH
- B) -COOH
- C) -CHO
- D) -NH₂ ✓

Answer: D

19. A saturated hydrocarbon contains:

- A) Only single bonds ✓
- B) Double bonds
- C) Triple bonds
- D) Both double and single bonds

Answer: A

20. An unsaturated hydrocarbon contains:

- A) Only single bonds
- B) Double or triple bonds ✓
- C) No bonds
- D) Only ionic bonds

Answer: B

HYDROCARBONS (ALKANES, ALKENES, ALKYNES) - 20 MCQs

1. Hydrocarbons are compounds containing:

- A) Carbon and oxygen
- B) Carbon and hydrogen ✓
- C) Carbon and nitrogen
- D) Hydrogen and oxygen

Answer: B

2. The general formula of alkanes is:

- A) C_nH_{2n}
- B) C_nH_{2n+2} ✓
- C) C_nH_{2n-2}

D) C_nH_{2n+1}

Answer: B

3. The general formula of alkenes is:

A) C_nH_{2n} ✓

B) C_nH_{2n+2}

C) C_nH_{2n-2}

D) C_nH_{2n+1}

Answer: A

4. The general formula of alkynes is:

A) C_nH_{2n}

B) C_nH_{2n+2}

C) C_nH_{2n-2} ✓

D) C_nH_{2n+1}

Answer: C

5. The first member of the alkane family is:

A) Methane ✓

B) Ethane

C) Propane

D) Butane

Answer: A

6. The first member of the alkene family is:

A) Methane

B) Ethene ✓

C) Propene

D) Butene

Answer: B

7. The first member of the alkyne family is:

- A) Methane
- B) Ethene
- C) Ethyne ✓
- D) Propyne

Answer: C

8. The formula of ethane is:

- A) CH_4
- B) C_2H_6 ✓
- C) C_2H_4
- D) C_2H_2

Answer: B

9. The formula of ethene is:

- A) CH_4
- B) C_2H_6
- C) C_2H_4 ✓
- D) C_2H_2

Answer: C

10. The formula of ethyne is:

- A) CH_4
- B) C_2H_6
- C) C_2H_4
- D) C_2H_2 ✓

Answer: D

11. The type of bond between carbon atoms in alkanes is:

- A) Single bond ✓

- B) Double bond
- C) Triple bond
- D) Ionic bond

Answer: A

12. The type of bond between carbon atoms in alkenes is:

- A) Single bond
- B) Double bond ✓
- C) Triple bond
- D) Ionic bond

Answer: B

13. The type of bond between carbon atoms in alkynes is:

- A) Single bond
- B) Double bond
- C) Triple bond ✓
- D) Ionic bond

Answer: C

14. Alkanes are also known as:

- A) Paraffins ✓
- B) Olefins
- C) Acetylenes
- D) Aromatics

Answer: A

15. Alkenes are also known as:

- A) Paraffins
- B) Olefins ✓
- C) Acetylenes
- D) Aromatics

Answer: B

16. Alkynes are also known as:

- A) Paraffins
- B) Olefins
- C) Acetylenes ✓
- D) Aromatics

Answer: C

17. The reaction where hydrogen is added to an unsaturated compound is called:

- A) Halogenation
- B) Hydrogenation ✓
- C) Hydration
- D) Combustion

Answer: B

18. Alkanes undergo mostly:

- A) Addition reactions
- B) Substitution reactions ✓
- C) Polymerization
- D) Hydration

Answer: B

19. Alkenes undergo mostly:

- A) Substitution reactions
- B) Addition reactions ✓
- C) Polymerization
- D) Combustion

Answer: B

20. The bromine water test is used to distinguish between:

- A) Alkanes and alkenes ✓
- B) Alkanes and alkynes
- C) Alkenes and alkynes
- D) None of these

Answer: A

BIOMOLECULES - 20 MCQs

1. Biomolecules are:

- A) Organic molecules found in living organisms ✓
- B) Inorganic molecules found in living organisms
- C) Synthetic molecules
- D) None of these

Answer: A

2. The four main types of biomolecules are:

- A) Carbohydrates, lipids, proteins, nucleic acids ✓
- B) Carbohydrates, fats, vitamins, minerals
- C) Proteins, nucleic acids, water, minerals
- D) None of these

Answer: A

3. Carbohydrates are made up of:

- A) Carbon, hydrogen, oxygen ✓
- B) Carbon, hydrogen, nitrogen
- C) Carbon, oxygen, nitrogen
- D) Hydrogen, oxygen, nitrogen

Answer: A

4. The building blocks of carbohydrates are:

- A) Amino acids
- B) Monosaccharides ✓
- C) Nucleotides
- D) Fatty acids

Answer: B

5. The building blocks of proteins are:

- A) Monosaccharides
- B) Amino acids ✓
- C) Nucleotides
- D) Fatty acids

Answer: B

6. The building blocks of nucleic acids are:

- A) Monosaccharides
- B) Amino acids
- C) Nucleotides ✓
- D) Fatty acids

Answer: C

7. The building blocks of lipids are:

- A) Monosaccharides
- B) Amino acids
- C) Nucleotides
- D) Fatty acids and glycerol ✓

Answer: D

8. The monomer of starch is:

- A) Amino acid

- B) Glucose ✓
- C) Nucleotide
- D) Fatty acid

Answer: B

9. The monomer of glycogen is:

- A) Amino acid
- B) Glucose ✓
- C) Nucleotide
- D) Fatty acid

Answer: B

10. The monomer of cellulose is:

- A) Amino acid
- B) Glucose ✓
- C) Nucleotide
- D) Fatty acid

Answer: B

11. The monomer of DNA is:

- A) Amino acid
- B) Glucose
- C) Nucleotide ✓
- D) Fatty acid

Answer: C

12. The monomer of RNA is:

- A) Amino acid
- B) Glucose
- C) Nucleotide ✓
- D) Fatty acid

Answer: C

13. The four nitrogenous bases in DNA are:

- A) Adenine, Guanine, Cytosine, Thymine ✓
- B) Adenine, Guanine, Cytosine, Uracil
- C) Adenine, Guanine, Thymine, Uracil
- D) None of these

Answer: A

14. The four nitrogenous bases in RNA are:

- A) Adenine, Guanine, Cytosine, Thymine
- B) Adenine, Guanine, Cytosine, Uracil ✓
- C) Adenine, Guanine, Thymine, Uracil
- D) None of these

Answer: B

15. The sugar found in DNA is:

- A) Ribose
- B) Deoxyribose ✓
- C) Glucose
- D) Sucrose

Answer: B

16. The sugar found in RNA is:

- A) Ribose ✓
- B) Deoxyribose
- C) Glucose
- D) Sucrose

Answer: A

17. The bond between amino acids is called:

- A) Glycosidic bond
- B) Peptide bond ✓
- C) Phosphodiester bond
- D) Hydrogen bond

Answer: B

18. The bond between nucleotides is called:

- A) Glycosidic bond
- B) Peptide bond
- C) Phosphodiester bond ✓
- D) Hydrogen bond

Answer: C

19. The bond between monosaccharides is called:

- A) Glycosidic bond ✓
- B) Peptide bond
- C) Phosphodiester bond
- D) Hydrogen bond

Answer: A

20. The most abundant biomolecule in living organisms is:

- A) Protein
- B) Lipid
- C) Carbohydrate
- D) Water ✓

Answer: D